Your Cancer Summary Report

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Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma with BRCA2 Germline Mutation

Precision Oncology Center — Demonstration Report

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⚠️ **DRAFT - FOR CLINICIAN REVIEW**

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Breast Cancer - ER+/PR+/HER2- Invasive Ductal Carcinoma

Stage: Stage IIA | **Grade:** Grade 2

You have Stage IIA breast cancer fuelled by estrogen and progesterone. This is an early-to-intermediate stage with strong treatment options. Genomic testing found two additional drug targets — BRCA2 and PIK3CA — opening further therapy choices.

✓ Positive Factors

- ER+/PR+ status — tamoxifen is active and effective
- BRCA2 germline mutation — eligible for FDA-approved PARP inhibitor (olaparib)
- PIK3CA H1047R — targetable with alpelisib + fulvestrant (FDA-approved)
- CCND1 amp + CDKN2A del — both support CDK4/6 inhibitor benefit
- ESR1 Moran's I 0.42 — hormone-driven spatial clustering confirmed
- Multi-omics integration: 12 samples, 1000 RNA / 400 protein / 250 phospho features; all 5 key genes FDR q < 7e-11

⚠ Challenges to Consider

- BRCA2 germline — 50% transmission risk; family genetic counselling essential
- MYC amplification may contribute to future treatment resistance

- Immune-cold TME — tumour has partially excluded immune cells

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA2

Pathogenic

c.5946delT frameshift, germline (AF=0.50, COSV57492880)

You were born with a change in the BRCA2 DNA-repair gene. This raised your cancer risk but makes the tumour sensitive to PARP inhibitor drugs that force cancer cells to self-destruct.

Treatment Option: Olaparib / Niraparib — FDA-approved for germline BRCA1/2-mutated HER2- breast cancer

PIK3CA

Pathogenic

H1047R missense, somatic (AF=0.42, COSM775)

An activating PIK3CA change is over-driving a key growth signal. An FDA-approved drug combination targets this directly.

Treatment Option: Alpelisib + fulvestrant — FDA-approved (SOLAR-1) for PIK3CA-mutated HR+/HER2- breast cancer

GATA3

Likely Pathogenic

Frameshift, somatic (AF=0.38, COSV104733902)

A GATA3 change confirms the hormone-driven identity of your cancer, commonly seen in ER+ breast cancer.

Treatment Option: No direct targeted therapy; reinforces continued endocrine strategy

MYC

Pathogenic — Copy Number Gain

Amplification chr8q24 (log2=1.45, ~5 copies)

Extra MYC copies are driving rapid cell growth. Investigational BET inhibitor drugs are being studied against this target.

Treatment Option: BET inhibitors (investigational) — consider clinical trial

CCND1

Pathogenic — Copy Number Gain

Amplification chr11q13 (log2=1.62, ~6 copies)

Extra CCND1 copies accelerate cell division — exactly the target of CDK4/6 inhibitors, standard-of-care additions to hormone therapy.

Treatment Option: Palbociclib / Ribociclib / Abemaciclib — FDA-approved for ER+/HER2- breast cancer

CDKN2A

Pathogenic — Copy Number Loss

Deletion chr9p21.3 (log2=−1.65, ~1 copy)

A missing CDKN2A braking gene lets the cell-division accelerator run unchecked. CDK4/6 inhibitors compensate for exactly this.

Treatment Option: Supports CDK4/6 inhibitor use; CDKN2A loss may predict enhanced response

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment
900-spot Visium across 7 regions. ESR1 mean 544 in tumour_core_luminal vs 17 in stroma (32-fold). CD8A/CD68 concentrated in stroma_immune — immune-excluded. ESR1 Moran's I=0.42 (clustered, z=264.7). MKI67 Moran's I=0.25 (heterogeneous).

Immune Status: ● Low immune activity - The tumor may be evading your immune system

Your tumour is strongly driven by oestrogen signalling in its centre. Immune cells sit around the edges rather than inside — confirming hormone-blocking therapy is on target.

Key Patterns Observed

- ESR1 32-fold enriched in tumour core — dominant luminal ER biology
- PGR (511) and GATA3 (665) co-localise in tumour core
- CD8A highest in stroma_immune (203) — immune-excluded
- MKI67 spatially heterogeneous (Moran's I 0.25)

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen + CDK4/6 Inhibitor (palbociclib / ribociclib / abemaciclib)

Endocrine + targeted

FDA Approved

Expected Response: 60-70% PFS improvement vs. endocrine alone

Your current tamoxifen blocks oestrogen. Adding a CDK4/6 inhibitor further slows cell division. CCND1 amp and CDKN2A del both predict strong benefit.

Why this may help you: ER+/PR+ biology plus CCND1 amplification and CDKN2A deletion

Common side effects: Neutropenia — monthly CBC required, Fatigue, Nausea, Hot flashes

Olaparib (PARP Inhibitor)

Targeted — PARP inhibitor

FDA Approved

Expected Response: 57-60% ORR (OlympiAD)

Olaparib exploits the BRCA2 repair defect — cancer cells accumulate fatal DNA damage while normal cells cope.

Why this may help you: Pathogenic germline BRCA2 frameshift — FDA-approved indication

Common side effects: Nausea, Fatigue, Anaemia, Decreased appetite

Alpelisib + Fulvestrant

Targeted — PI3Kα inhibitor + endocrine

FDA Approved

Expected Response: 35-40% PFS benefit (SOLAR-1)

Alpelisib blocks the PIK3CA H1047R growth signal. Combined with fulvestrant for a two-pronged attack.

Why this may help you: PIK3CA H1047R at 42% allele frequency — the SOLAR-1 FDA approval mutation

Common side effects: Hyperglycaemia — close blood-sugar monitoring, Rash, Diarrhoea, Nausea

Clinical Trials

You may be eligible for these research studies offering new treatments:

OlympiA: Adjuvant Olaparib in Germline BRCA-Mutated HER2-Negative Breast Cancer
Phase: 3 | **Status:** Active

Trial ID: NCT03796040

Adjuvant olaparib for germline BRCA2 HER2-negative breast cancer — your exact profile.

Contact: clinicaltrials.gov/ct2/show/NCT03796040

INAVO120: Inavolisib + Palbociclib + Fulvestrant in PIK3CA-Mutated HR+/HER2- Breast Cancer
Phase: 3 | **Status:** Active

Trial ID: NCT05242367

Next-generation PI3K inhibitor targeting your PIK3CA H1047R mutation combined with CDK4/6 inhibitor and fulvestrant.

Contact: clinicaltrials.gov/ct2/show/NCT05242367

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 6 months	Breast MRI or mammogram	Monitor tumour response and detect new lesions
Every 3 months	CEA and CA 15-3	Current levels normal (CEA 2.1, CA 15-3 18); rising values signal progression
Monthly on CDK4/6 or PARP inhibitor	CBC	Detect bone-marrow suppression
Monthly if starting alpelisib	HbA1c / blood glucose	Alpelisib commonly causes high blood sugar
Annually	Pelvic ultrasound + gynaecologic exam	BRCA2 germline increases ovarian cancer risk
Within 30 days	Genetic counselling referral	50% transmission risk to first-degree relatives

⚠️ When to Call Your Doctor

Contact your care team right away if you experience:

- New or worsening bone pain
- Shortness of breath or chest pain
- Fever above 38°C on CDK4/6 inhibitor
- Unusual bruising or bleeding
- Extreme fatigue or pallor

Questions? Oncology nurse navigator or 24-hour triage line

For Your Family

The BRCA2 mutation is inherited. Each child has a 50% chance of inheriting it; siblings and parents have 50% risk. Carriers face 45-85% lifetime breast cancer risk and 10-20% ovarian cancer risk.

Recommended Steps for Family Members

- Schedule genetic counselling within 4 weeks
- Inform first-degree relatives and recommend testing
- Discuss risk-reducing surgery with your gynaecologist
- Consider referral to a hereditary cancer programme

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:

📣 **BRCA Foundation**

Education, support, and genetic testing resources for BRCA1/2 carriers and families

📞 1-800-292-2776 | 🌐 https://brcafoundation.org

👥 **Susan G. Komen Helpline**

Breast cancer helpline, financial assistance, and peer community support

📞 1-877-465-6636 | 🌐 https://www.komen.org

🧬 **FORCE — Facing Our Risk of Cancer Empowered**

Dedicated hereditary breast and ovarian cancer support; BRCA-specific peer navigation

| 🌐 https://www.facingourrisk.org

⚠️ **Important Notice**

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. Requires oncologist review before any treatment decisions.

Data Sources: mockepic (DRY_RUN), fgbio (DRY_RUN), genomic-results (fixture files), spatialtools (Visium 900-spot), multiomics (12 samples, 1650 features — Fix 5 ✅), resolve_patient_data_paths (Fix 2 ✅), set_patient_context (Fix 7 ✅)
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